Youssef Fellah

Software Engineer & Systems Architect

Backend · Distributed Systems · AI & Agent Orchestration



Seeking a Final-Year Internship (PFE) — available from January / February 2027 — 4 to 6 months

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ABOUT

Software Engineering student at Université Mundiapolis, specializing in **backend development (Java / Spring Boot)** and **systems architecture** (Clean Architecture, DDD, UML 2.5, BPMN 2.0). I design secure, modular, and scalable distributed platforms, backed by hands-on experience in multi-tenant SaaS, ERP engineering, RAG pipelines, and autonomous agent orchestration.

I am looking for a **Final-Year Internship (PFE) starting in January – February 2027**, where I can contribute to the design, modeling, and development of high-concurrency distributed systems. Rigorous and autonomous, I take full ownership from formal analysis to production deployment.

TECHNICAL SKILLS

Self-assessed proficiency, out of 5.

Backend & Microservices	Java, Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST / RPC APIs	●●●●○
Architecture & Design	Clean Architecture, DDD, SOLID, Design Patterns (GoF), UML 2.5, BPMN 2.0, ADRs	●●●●○
Project Delivery	Agile / Scrum, Git workflows, rigorous technical documentation	●●●●○
Databases	PostgreSQL, MySQL, SQL tuning, Redis, MongoDB, vector search	●●●●○
Cloud & DevOps	Docker, Linux, GitHub Actions (CI/CD), AWS (EC2, S3, RDS)	●●●○○
Security & Networks	RBAC / IAM modeling, stateless JWT auth, OWASP, OPNsense, VLANs (802.1Q)	●●●○○
AI & Orchestration	RAG pipelines, LLM APIs, Agent Orchestration	●●●○○
Scripting & Languages	Python, Bash, TypeScript, JavaScript, PHP, C, Solidity	●●●○○
Frontend	Angular, React.js, Next.js	●●●○○

PERSONAL SKILLS

Teamwork — collaborative, mentors peers, shared standards

Autonomy — drives a solution from analysis to deployment

Rigor — maintainability, documentation, quality

Fast learner — adopts new stacks and domains quickly

Critical thinking — analyzes constraints before building

Communication — turns business needs into clear specs

EXPERIENCE

Feb. 2026 – Aug. 2026
Casablanca, Morocco

Spring Boot · Angular · PostgreSQL · JWT / RBAC · RAG

Software Architect & Systems Engineer (Work-Study)

Broker Immobilier

- Architected a multi-tenant SaaS real estate CRM with Spring Boot, applying Clean Architecture to isolate the domain layer and ensure long-term maintainability.
- Enforced end-to-end security with stateless JWT authentication and granular Role-Based Access Control (RBAC).
- Integrated a RAG pipeline and LLM APIs for semantic search, document summarization, and metadata extraction.

Aug. 2025 – Oct. 2025
Casablanca, Morocco
PHP · Dolibarr · MySQL · SQL optimization

Information Systems & ERP Engineering Intern
HTBS Africa

- Translated Moroccan tax and treasury compliance rules into formal technical specifications for Dolibarr ERP.
- Developed and deployed 4 custom modular extensions (Treasury, Invoicing, Tax Reporting) following the Open-Closed Principle, without touching the application core.
- Optimized database schemas and refactored bottleneck SQL queries, reducing financial report generation time by 40%.

Apr. 2025 – Jul. 2025
Errachidia, Morocco
Solidity · Ethereum · IPFS · React.js

Blockchain Developer Intern (Bachelor's Final Project)
FST Errachidia

- Designed and deployed an Ethereum-based platform for decentralized e-learning and credential verification.
- Wrote and audited Solidity smart contracts for tamper-proof diploma issuance; integrated IPFS for decentralized file storage.

PROJECTS

Java · Spring Boot · PostgreSQL

Core Banking Transaction Engine
github.com/ind4x/spring-boot-portfolio
Decoupled architecture isolating the transactional core from external client APIs, with strict ACID boundaries, idempotent processing, and RBAC authorization over an integration bus.

Python · Docker · Bash

devctl — Platform Engineering CLI
github.com/ind4x/devctl
Asynchronous CLI automating environment setup, Docker lifecycle, and dependency orchestration across Spring Boot, Angular, and Vue.js stacks, removing cross-environment configuration drift.

AWS · Zabbix · OPNsense

AWS Supervision & OPNsense Perimeter
github.com/ind4x/aws-zabbix-monitoring
Zabbix monitoring on AWS EC2 for Linux and Windows Server; OPNsense firewall with VLAN (802.1Q) segmentation and network hardening.

BPMN 2.0 · Python · Odoo 17

Logistics Modeling & Odoo Modules
github.com/ind4x/si-portfolio
Perishable-goods logistics flows modeled in BPMN 2.0; Odoo modules built on the Open-Closed principle for full traceability.

Git · GitHub Actions · ADRs

Software Quality & Engineering Governance
github.com/ayaaas/projet-git-groupe-d
Defined branching strategies and CI validation policies, wrote Architecture Decision Records (ADRs), and mentored fellow developers.

EDUCATION

2025 – Present
Casablanca, Morocco

Engineering Degree (Diplôme d'Ingénieur) in Software Engineering
Université Mundiapolis
Distributed Systems, Advanced Software Architecture, Enterprise Frameworks, Cloud Infrastructure

2024 – 2025
Errachidia, Morocco

Bachelor of Science and Technology (LST) in Software Engineering
FST Errachidia
Algorithms & Data Structures, Relational Databases, Operating Systems, Web & Mobile Engineering

CERTIFICATIONS

- **AWS Academy Graduate — Cloud Foundations** · Amazon Web Services, 2025
- **Advanced Java APIs and Functional Programming** · Packt / Coursera, 2026
- **HCIA-Cloud Computing V5.5** · Huawei ICT Academy, 2025
- **GitHub Foundations** · Microsoft / GitHub, 2026

LANGUAGES

Arabic — Native

French — Fluent, full professional

English — Professional working